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Causal SCX: Causal Inference

Causal SCX — Multi-Expert Audit of Causal Inference

SCX

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Abstract

SCX_{scx_causal}, 3-SCM (?) Disagreement $M \geq 2$ (1) **Causal Consensus****Causal Consensus Bound:** Hoeffding awaiting, $M \{\hat{\tau}_m\}_{m=1}^M$ ConsistencyATE τ^* .Proof M, τ^* , **systematic confounding** (2) $Z_1, Z_2, \dots, Z_M$ $M U T Y Z_m, D(\hat{\tau}_1, \dots, \hat{\tau}_M)$. (3) $\{Z_1, \dots, Z_K\}$, IV Disagreement Q_k k . (4) **do-calculus** do Disagreement., SCX **Key-words:** do-calculus,

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—, , awaiting. **SCX**_{scx_causal}(?) Disagreement .Core—**3-SCM** Theorem 1, ?—SCM $M > 3$ **Causal SCX**—Causal Inference. do-calculus SCX . : $T \in \{0, 1\}, Y \in \mathbb{R}, X \in \mathbb{R}^p$. ATE $\tau^* = \mathbb{E}[Y(1) - Y(0)], Y(t)$. $M \hat{\tau}_1, \dots, \hat{\tau}_M, \mathcal{M}_m$.

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2.1

D 1 (SCM). *SCM* $\mathcal{M} = \langle U, V, F, P(U) \rangle$ $U, V, F = \{f_v : v \in V\} U$. $\mathcal{M} P_{\mathcal{M}}(V) P_{\mathcal{M}}(V | \text{do}(t))$.

D 2 ($\mathcal{M}$ -awaiting). *SCM* $\mathcal{M}_1, \mathcal{M}_2$ awaiting, $\mathcal{M}_1 \equiv_{obs} \mathcal{M}_2$, : $P_{\mathcal{M}_1}(V) = P_{\mathcal{M}_2}(V)$.

Theorem 1 (3-SCM): awaiting SCM $\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3, Q$ ATE:

$$Q(\mathcal{M}_1) \neq Q(\mathcal{M}_2) \neq Q(\mathcal{M}_3) \neq Q(\mathcal{M}_1),$$

Q . M , Disagreement.

2.2 Hoeffding

Hoeffding awaiting: $X_1, \dots, X_M, a_i \leq X_i \leq b_i, \varepsilon > 0$,

$$P \left(\left| \frac{1}{M} \sum_{i=1}^M X_i - \mathbb{E} \left[\frac{1}{M} \sum_{i=1}^M X_i \right] \right| \geq \varepsilon \right) \leq 2 \exp \left(- \frac{2M^2 \varepsilon^2}{\sum_{i=1}^M (b_i - a_i)^2} \right).$$

Core: — $P(V)$, — $\hat{\tau}_m$

3 Causal Consensus

3.1

$M \{ \hat{\tau}_m \}_{m=1}^M \mathcal{Z}_m, g_m \mathcal{H}_m \text{ ATE} :$

$$\hat{\tau}_m = \hat{\tau}(\mathcal{Z}_m; g_m; \mathcal{H}_m).$$

consensus measure:

$$C_M = 1 - \frac{1}{M(M-1)} \sum_{1 \leq i < j \leq M} |\hat{\tau}_i - \hat{\tau}_j|.$$

$C_M = 1$; Disagreement $C_M \rightarrow 0$.

3.2

$$\hat{\tau}_m - \tau^* = \underbrace{\text{bias}(\mathcal{Z}_m, \mathcal{H}_m)} + \underbrace{\varepsilon_m}.$$

D 3. *consensus offset:*

$$\delta_M = \frac{1}{M} \sum_{m=1}^M \text{bias}(\mathcal{Z}_m, \mathcal{H}_m).$$

3.3

T 4 (Causal Consensus — Hoeffding). (A1) $\hat{\tau}_m \in [\tau_{\min}, \tau_{\max}]$;

(A2) $\varepsilon_m \in P(V)$ /

(A3) $m_0 \text{ bias}(\mathcal{Z}_{m_0}, \mathcal{H}_{m_0}) = 0$
 $\varepsilon > 0$:

$$P \left(\left| \frac{1}{M} \sum_{m=1}^M \hat{\tau}_m - \tau^* - \delta_M \right| \geq \varepsilon \right) \leq 2 \exp \left(- \frac{2M\varepsilon^2}{(\tau_{\max} - \tau_{\min})^2} \right).$$

, $\delta_M = 0$, $O(1/\sqrt{M})$ τ^* , ATE, δ_M τ^* Scale, $C_M \rightarrow 1$.

Proof Sketch. $\hat{\tau}_m \in \tau^* + \text{bias}_m + \varepsilon_m$. $\delta_M + \frac{1}{M} \sum_m \varepsilon_m$. (A2) , $\varepsilon_m \in P(V)$, Hoeffding Conclusion.
 Core: Hoeffding δ_M — “Causal Inference” awaiting”. $\square$ $\square$

Honest Strike / Honest Critique:

Honest Strike: τ^* . **Honest Strike:** τ^* , τ^* , τ^* .

C 5. τ^* , $\delta_M \neq 0$ $|\delta_M| > \frac{\tau_{\max} - \tau_{\min}}{\sqrt{2M}}$, $\{ \hat{\tau}_m \}_{m=1}^M$.

3.4

, M , consensus depth:

$$D(M) = \inf \{ d \in \mathbb{N} : \exists d \}.$$

$D(M) > M/2$, — ..

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4.1

$M \mathcal{Z}_1, \dots, \mathcal{Z}_M \subseteq X. \mathcal{Z}^*, : \mathcal{Z}^* T Y. \mathcal{Z}_m \subsetneq \mathcal{Z}^*, \hat{\tau}_m; \mathcal{Z}_m \supseteq \mathcal{Z}^*, .$

A 6. $: \mathcal{Z}_1 \preceq \mathcal{Z}_2 \iff \mathcal{Z}_1 \subseteq \mathcal{Z}_2. \mathcal{Z}_{\min}^*.$

T 7. Disagreement:

$$\Sigma_{dis} = (|\hat{\tau}_i - \hat{\tau}_j|)_{i,j=1}^M \in \mathbb{R}^{M \times M},$$

$\rho(\Sigma_{dis}). :$

$$G_m = \frac{\hat{\tau}_m - \hat{\tau}_{m-1}}{|\mathcal{Z}_m| - |\mathcal{Z}_{m-1}|},$$

$|\mathcal{Z}_m| \mid |\mathcal{Z}_1| < |\mathcal{Z}_2| < \dots. \mathcal{H}_0:$

(i) $\rho(\Sigma_{dis}) \xrightarrow{p} 0$ as $n \rightarrow \infty$;

(ii) $\sup_m |G_m| \xrightarrow{p} 0$ as $n \rightarrow \infty$;

(iii) $\hat{\tau}_m : \exists m_0 : \forall m \geq m_0, |\hat{\tau}_m - \tau^*| < |\hat{\tau}_{m-1} - \tau^*|.$

(i) (ii) .

Proof Sketch. $\mathcal{Z}_m \mathcal{Z}_m \supseteq \mathcal{Z}_{\min}^*, \hat{\tau}_m \tau^*: \hat{\tau}_m \xrightarrow{p} \tau^*, \forall m. \Sigma_{dis} \xrightarrow{p} 0 G_m \xrightarrow{p} 0. , \mathcal{Z}_m \not\supseteq \mathcal{Z}_{\min}^*, , \hat{\tau}_m \xrightarrow{p} \tau^* + b_m b_m \neq 0. , b_m \Sigma_{dis} . , \text{Appendix.} \quad \square \quad \square$

D 8.

$$D_{conf} = \frac{\rho(\Sigma_{dis})}{\frac{1}{M} \sum_{m=1}^M \widehat{\text{Var}}(\hat{\tau}_m)},$$

$\widehat{\text{Var}}(\hat{\tau}_m) \hat{\tau}_m . \mathcal{H}_0 , D_{conf} \text{ Tracy-Widom} .$

”Missing”””” . , $\rho(\Sigma_{dis})$ **Honest Strike:** ———

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5.1

$K \geq 2 \mathcal{Z}_1, \dots, \mathcal{Z}_K, \mathcal{Z}_k :$

(IV1) **Relevance:** $\mathbb{C}\text{ov}(\mathcal{Z}_k, T) \neq 0$;

(IV2) **Exclusion:** $\mathcal{Z}_k \perp\!\!\!\perp Y(t) \mid T, X$;

(IV3) **Ignorability:** $\mathcal{Z}_k \perp\!\!\!\perp U.$

$\hat{\tau}_{\text{IV}}^{(k)} \mathcal{Z}_k . \hat{\tau}_{\text{IV}}^{\text{full}} K , \hat{\tau}_{\text{IV}}^{(-k)} \{Z_1, \dots, Z_K\} \setminus \{Z_k\} .$

5.2

T 9 (IV). k :

$$Q_k = \frac{\hat{\tau}_{IV}^{full} - \hat{\tau}_{IV}^{(-k)}}{\sqrt{\widehat{\text{Var}}(\hat{\tau}_{IV}^{full} - \hat{\tau}_{IV}^{(-k)})}}.$$

Z_k (IV1)–(IV3) Z_k ,

$$Q_k \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{as } n \rightarrow \infty.$$

Z_k (IV2), $Z_k \rightarrow Y$, $\gamma_k \neq 0$,

$$Q_k \xrightarrow{d} \mathcal{N}\left(\frac{\gamma_k}{\sqrt{\widehat{\text{Var}}(\dots)}}, 1\right),$$

γ_k ScaleGrowth.

Proof Sketch. IV , $\hat{\tau}_{IV}^{full} \hat{\tau}_{IV}^{(-k)} \tau^*$. Z_k , $\hat{\tau}_{IV}^{full} \hat{\tau}_{IV}^{(-k)}$, Z_k . Delta . □ □

C 10.

$$Q_{\max} = \max_{k=1, \dots, K} |Q_k|.$$

””, $Q_{\max} K Q_{\max}$, .

5.3 Disagreement

M . m Verdict $\mathbf{v}_m \in \{0, 1\}^K$, $v_{m,k} = 1$ Z_k . **Disagreement:**

$$D_{IV} = \frac{1}{M(M-1)} \sum_{i < j} \|\mathbf{v}_i - \mathbf{v}_j\|_1.$$

D_{IV} Disagreement. $D_{IV} Q_k k$; $D_{IV} Q_k$.

Honest Strike / Honest Critique:

Honest Strike: IV $F < 10$, , Q_k **Honest Strike:** . Q_k Consist

6 do-calculus

6.1 Pearl

G DAG, X, Y, Z, W . do-calculus (?):

- **1/:** $P(y \mid \text{do}(x), z, w) = P(y \mid \text{do}(x), w) \ G_{\overline{X}} Y \perp\!\!\!\perp Z \mid X, W$.
- **2:** $P(y \mid \text{do}(x), \text{do}(z), w) = P(y \mid \text{do}(x), z, w) \ G_{\overline{X}\underline{Z}} Y \perp\!\!\!\perp Z \mid X, W$.
- **3/:** $P(y \mid \text{do}(x), \text{do}(z), w) = P(y \mid \text{do}(x), w) \ G_{\overline{XZ(W)}} Y \perp\!\!\!\perp Z \mid X, W$.

6.2

T 11 (do-calculus-SCX). $\mathcal{E} = \{\mathcal{M}_1, \dots, \mathcal{M}_M\}$ M , $\mathcal{Q}$.

(1) $1 \ i, j \ \mathcal{Z}_i \neq \mathcal{Z}_j \text{ awaiting } \hat{\tau}_i \approx \hat{\tau}_j, 1 \ G_{\overline{T}} \text{ --- } \mathcal{Z}_i \Delta \mathcal{Z}_j$

(2) $2 \ G_{\overline{XZ}} \text{ --- } . \text{ scx_causal Theorem 1}$

(3) $3 \ Z \text{ do-Disagreement, SCX } D_{IV} \text{ Disagreement } G_{\overline{XZ(W)}} .$

: **SCX**

Proof Sketch. (1) $(\Rightarrow)$: do $r \Rightarrow$ (2) $(\Leftarrow)$: Disagreement SCX $\Rightarrow$.Disagreement, Core Pearl Bareinboim $\text{ --- do-calculus (??) --- scx_causal .}$ $\square$ $\square$

C 12 (Disagreement do). *SCX M Disagreement $\nexists$ Consistency $n \rightarrow \infty$, do DAG ---*

Honest Strike / Honest Critique:

Honest Strike: . scx_causal 3-SCM Core: **awaitingDisagreement.** **Honest Strike:** do-calculus (?). DAG ; , c -awaiting. **Honest Strike:** ?? Proof”

7

7.1 M

$M \rightarrow \infty$, $\mathcal{H} \Pi$, :

$$\bar{\tau}_{\infty} = \int_{\mathcal{M} \in \mathcal{H}} \tau(\mathcal{M}) d\Pi(\mathcal{M}),$$

ATE Π -. τ^* , Π , $\bar{\tau}_{\infty}$. SCX Causal Inference (?).

7.2 Compactness

?? : M , $M-1 \rightarrow \max, 1$.Hoeffding $\text{ --- } M-1$ Compactness $R \in \mathbb{R}^{M \times M}$, $R_{ij} = \text{Corr}(\hat{\tau}_i, \hat{\tau}_j)$.

7.3

1 : C_M , ;

2 : D_{conf} , ;

3 : IV , $\{Q_k\}_{k=1}^K$;

4 **do** Disagreement, ??

5 **Verdict:** , ATE

7.4

(1) : .

(2) : M

(3) : $p \gg n$,

(4) : . , SCX g -

(5) :

8 Conclusion

Causal SCX—Causal Inference, scx_causal 3-SCM M . Proof do-calculus SCX. Core: , Causal Inference— M , . ——— Causal SCX , M , ——— **Consistency** δ_M . Conclusion.: , Causal Inference””

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A ?? Proof

Proof. : $\mathcal{Z}_m \supseteq \mathcal{Z}_{\min}^*$, Pearl, 2009, Theorem 3.3.2, $\mathcal{Z}_m : \hat{\tau}_m \xrightarrow{p} \tau^*$ as $n \rightarrow \infty$. $i, j: |\hat{\tau}_i - \hat{\tau}_j| \xrightarrow{p} 0$. Σ_{dis} , , $\rho(\Sigma_{\text{dis}}) \xrightarrow{p} 0$. , $|\mathcal{Z}_m| \geq |\mathcal{Z}_{\min}^*|$, $G_m \xrightarrow{p} 0$. : $\mathcal{Z}_m \not\supseteq \mathcal{Z}_{\min}^*$, $T \leftarrow U \rightarrow Y$ $\mathcal{Z}_m$. Imbens & Rubin, 2015, Ch.12:

$$\hat{\tau}_m \xrightarrow{p} \tau^* + b_m, \quad b_m \neq 0,$$

b_m ScaleMissing. $S_{\text{conf}} = \{m : \mathcal{Z}_m \not\supseteq \mathcal{Z}_{\min}^*\}$. $i \in S_{\text{conf}}, j \notin S_{\text{conf}}: |\hat{\tau}_i - \hat{\tau}_j| \xrightarrow{p} |b_i| > 0$. Gershgorin , $|S_{\text{conf}}| \geq 1$ $\rho(\Sigma_{\text{dis}}) \min_{i \in S_{\text{conf}}} |b_i|$. $\therefore \Sigma_{\text{dis}}$. $\mathcal{H}_0$, $O_p(n^{-1/2})$, Random MatrixTracy–Widom, 1996, F_β $\beta = 1$. $D_{\text{conf}} \mathcal{H}_0$ Tracy–Widom . $\square$ $\square$

B ?? Delta

Proof. $T = \alpha_0 + \mathbf{Z}'\boldsymbol{\alpha}_1 + \mathbf{X}'\boldsymbol{\alpha}_2 + \varepsilon$. $Y = \beta_0 + \hat{T}\beta_1 + \mathbf{X}'\boldsymbol{\beta}_2 + \eta$, $\hat{\tau}_{IV} = \hat{\beta}_1$ ATE IV . , $\sqrt{n}(\hat{\tau}_{IV} - \tau^*) \xrightarrow{d} \mathcal{N}(0, \sigma_{IV}^2)$. $\hat{\tau}_{IV}^{\text{full}} - \hat{\tau}_{IV}^{(-k)}$. $\hat{\tau}_{IV}^{\text{full}} - \hat{\tau}_{IV}^{(-k)}$ influence function:

$$\psi_k(O) = \psi^{\text{full}}(O) - \psi^{(-k)}(O),$$

$\psi^{(\cdot)}$. , $\mathbb{E}[\psi_k(O)] = 0$, $Q_k \xrightarrow{d} \mathcal{N}(0, 1)$. Z_k , $\mathbb{E}[\psi^{\text{full}}(O)] = \gamma_k \cdot h(\cdot)$, $\mathbb{E}[\psi^{(-k)}(O)] = 0$, $\lambda_k(n) = O(\sqrt{n}\gamma_k)$. □